# Manual

## Processing of Alternative / Parallel Sequences

## BDE-APF 8.1

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# Contents

# 1 Overview Processing Alternative / Parallel Sequences

## Purpose

## Implementation considerations

You use the function package if:

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## Integration

## Features

- Processing of alternative or parallel sequences
  - Supplementary function for processing alternative or parallel sequences in HYDRA
  - Configuration option used to activate sequence-based functions in HYDRA as part of customizing.
  - Administration of standard sequence and alternative or parallel sequences
  - Option of transferring alternative sequences (customer-specific interface)
  - Tabular sequence display with option to replace sequences from alternative sequences in HYDRA to short-term production control (uploading the sequence replacement to SAP requires that a customer-specific interface has been realized)
  - Maintenance or processing dialog used to display or create order sequences
  - Consideration of the smallest quantity of a parallel sequence during an active plausibility check for send-ahead quantity.
  - Transfer of parallel sequences from SAP PP via the PP-PDC interface
  - Consideration of sequences during time ticket-based upload via PP-PDC interface to SAP PP.

  Please note: Adding and coordinating the specific requirements and implementing them are considered a customized HYDRA service (a service subject to an added charge).

# 2 Edit Order Sequences

## Summary

| | |
|---|---|
| Menu | Order Management → Order Management → Edit order sequences |
| Transaction code | edseq |
| Function authorization | edseq |

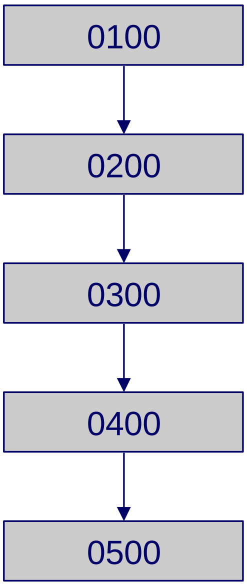
## Usage

Operations within an order are grouped into sequences in order to create a summary of them. Production uses this information as an orientation tool to process each operation. Within the sequence, the operations are processed in sequence one at a time. By linking several sequences within the order, network-type structures can be illustrated.

You also have the option to use parallel or alternative order sequences. The following sequence types are supported:

**Standard sequence**

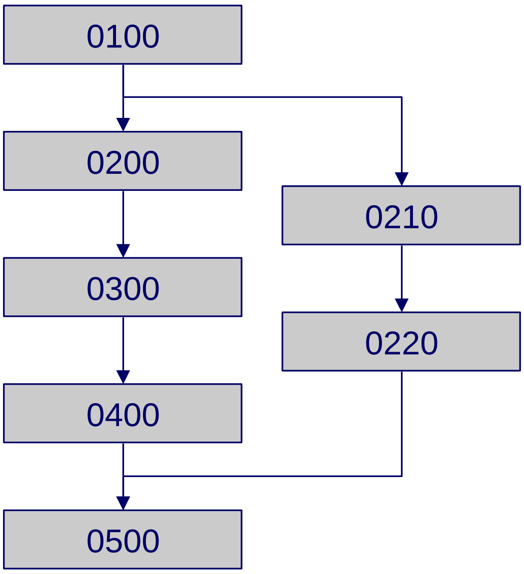
The standard sequence is available by default and describes the first sequence of the order.




For a purely sequential order, only the standard sequence is required. If certain operations are to be processed in parallel or alternatively to the standard sequence, they must be grouped in relevant sequences. Thus, parallel and alternative sequences can branch off of a single standard sequence.

**Parallel sequences**

A parallel sequence runs parallel to a partial sequence of the standard sequence. It is used, for example, when certain processes are to run at the same time (in parallel). This may be the case, for example, in the processing industry.



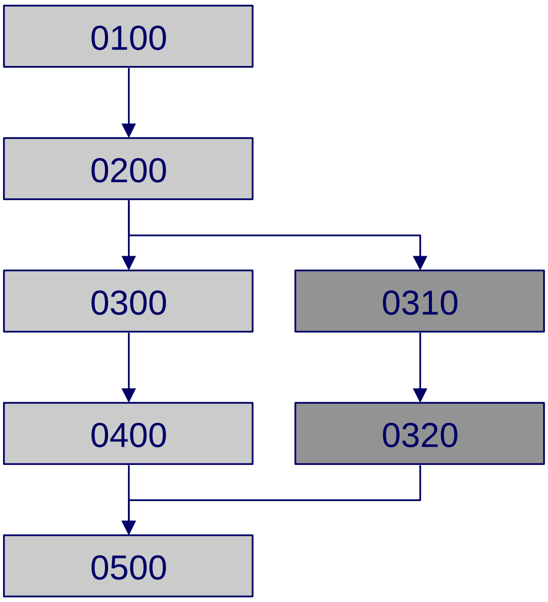

This partial sequence is defined by the branch operation and the return operation of this particular reference sequence. As such, the start of the parallel sequence is equal to the start of the branch operation in the reference sequence and the end is equal to the end of the reference sequence's return operation.

**Alternative sequences**

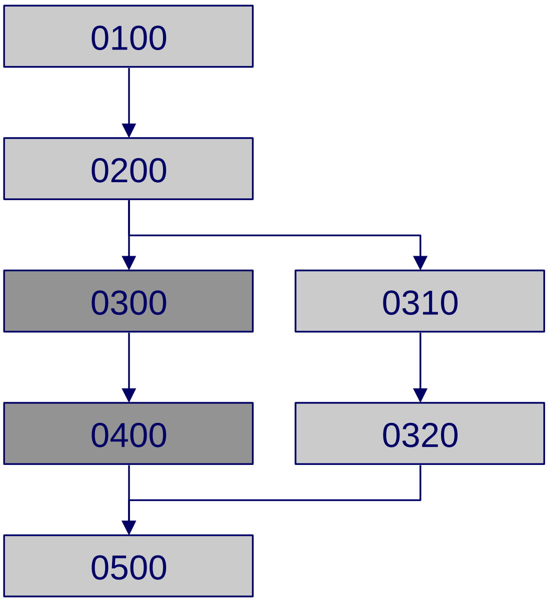
An alternative sequence describes one or more operations, which can be used alternatively to a partial sequence of the standard sequence. It is used, for example, if the production process varies for certain batch sizes.

Alternative sequences each have one active sequence that is relevant for processing.

Order with an inactive, alternative sequence




Order with an active, alternative sequence



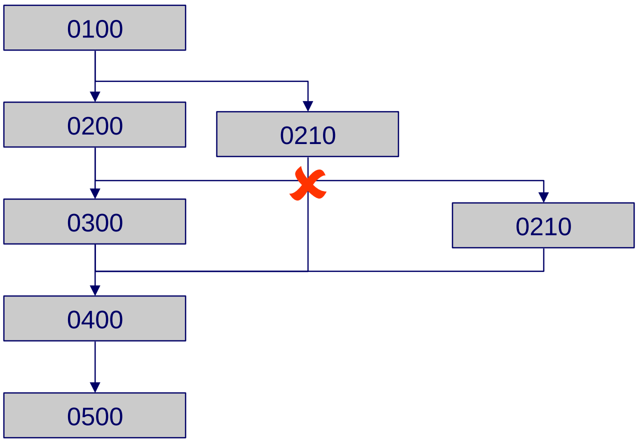

Inactive sequences are not considered either in scheduling or in detailed planning; they can neither be posted.

It is possible to activate an alternative sequence at the HYDRA console if certain conditions are met. This occurs interactively.

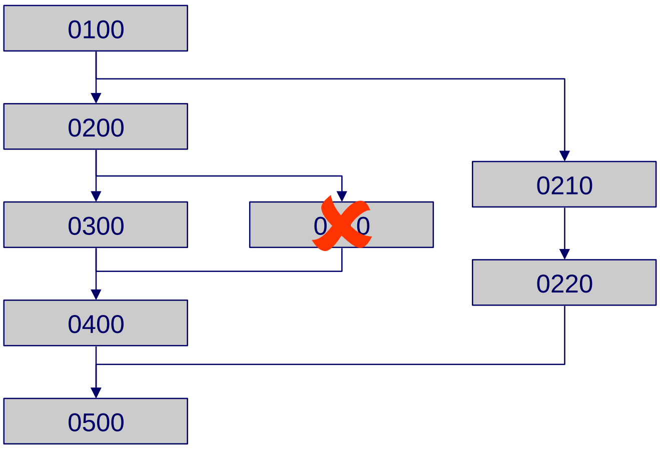
## General

- Each operation is assigned to exactly one sequence.
- The relationships that are created as a result between the operations and between the sequences are stored by HYDRA in an internal relationship table.
- Parallel sequences always run parallel to the standard sequence.
- Alternative sequences always run alternately to the standard sequence.

- Operation sequences of different sequences may not overlap.




- With respect to a partial sequence of the standard sequence, which is restricted by a branch OP and a return OP of a parallel or alternative sequence, there may be no parallel or alternative sequence with a branch operation and/or return operation within it.




## Deleting sequences

- A sequence may only be deleted if there is no operation that is assigned to this sequence.
- As a rule, a standard sequence cannot be deleted.

## Creating and deleting operations

- When an operation is created or deleted, HYDRA automatically updates the order network for this order. The order network documents the relationships between operations and this information is used for planning in HYDRA shop floor scheduling (HLS) as well as for processing/ posting.

## Copying an order

- If an order is copied, the new order is available in its "initial state". This means that any existing alternative sequences are generally inactive, even if they were previously active in the order that was copied.

## Operation status

- It cannot be determined based on the operation status whether the operation is a part of the standard sequence's active alternative sequence or its inactive alternative sequence or rather its inactive partial sequence (as a result of activating an alternative sequence), because the operation status does not change during activation or deactivation.
- Operations of an inactive alternative sequence have the same initial status *prepared* when they are newly created, just as is the case for operations of active sequences.

## Sequencing list

- Operations of the standard sequence's inactive alternative sequence or an inactive partial sequence (as a result of activating an alternative sequence) are not shown in the sequencing list. They can neither be logged on.

## Merged operation

- Operations of the standard sequence's inactive alternative sequence or inactive partial sequence (as a result of activating an alternative sequence) cannot be a part of a merged operation.

## Issues of note relating to parallel sequences

- Parallel sequence structures are accounted for in the planning algorithms in HYDRA shop floor scheduling (HLS)

- If the option "Checking status of preceding OP" is active (customized HYDRA feature), after merging several parallel sequences, an operation may not be logged on until all sequences or rather their last operations have been interrupted or have been finished (depending on the customized settings). Any overlapping is considered in the process.

- The smallest yield of the consolidated sequences, or rather of their last operations, is considered the send-ahead quantity of several parallel sequences. This is carried forward as the target quantity to the successors if the processing code provides for a target quantity update at the operation (customized HYDRA feature).

## Integration

Please note the following with regard to displaying operations of alternative sequences in the MOC functions and evaluations/reports:

**Operations/ operations logged on/ pool of orders**

An operation of an inactive sequence ("inactive operation") can be recognized by the "Y" in the column *Control.*

If no operations of inactive sequences are to be displayed in the order overview, then all of the options except for the option "Y" must be set in the selection range in the *Control* selection field.

**Order overview**

In the *Progress* index tab, operations of both active as well as inactive sequences are displayed.

**Order information**

In the order information, operations of both active as well as inactive sequences are displayed. In order to be able to recognize inactive sequences as such, you use the column configurator in the operation table to have the *control* column displayed. In this column, operations of inactive sequences are listed with a "Y".

## Requirement

In order to process sequences in HYDRA, the relevant license must have been issued. It is not possible to use DOS based terminals.

The following activities are required for use:

1. The sequence number length must have been set in the basic HYDRA settings

> **WARNING**
> You may only set the sequence number length during the initial HYDRA setup process and provided that no order backlog data exists in HYDRA. Any subsequent settings or changes will make the system act inconsistently.

2. Reactivating dynamic dialogs
   As a result, the input fields at the Windows terminal will expand by the defined sequence number length.

## Selection criteria

The application provides the following selection criteria:

**Order**

Enter the order number for the order with the sequences you would like to display.

## Field descriptions

**Order number**

Order for which the sequence is defined. A sequence can only be set up for an order, if the order header already exists in HYDRA.

**Sequence**

Identification of the sequence within an order.

Please note: The standard sequence always has the sequence number 0.

If the "sequence" field is not shown in the editing dialog, the sequence number length is 0 in the basic parameter settings. Please contact MPDV.

**Designation**

Description of the sequence.

**Sequence category**

**S** = Standard sequence

There is only one standard sequence for every order; the standard sequence cannot be deleted.

**P** = Parallel sequence to the standard sequence

There can be several parallel sequences for each order

**A** = Alternative sequence to the standard sequence

There can be several alternative sequences for each order

**Please note:**

The sequence category cannot be edited after a sequence has been set up!

**Active**

The qualification "Active" is only relevant for alternative sequences:

J = Active

N = Not active

If a new alternative sequence is set up, it is set as *not active.*

For standard sequences and alternative sequences, this qualification is always set to *active.*

**Orientation**

If there are several parallel sequences, the lead times generally vary in length. This creates time buffers in the sequences. The orientation function controls whether these buffers are at the beginning or the end of the sequence. The following options are available:

F = Earliest due date
If the sequence is set for the earliest date, the buffer will be at the end of the sequence.

S = Latest due date
If the sequence is set for the latest date, the buffer will be at the beginning of the sequence.

N = Not relevant; this is the case for standard sequences and alternative sequences.

If there are several parallel sequences for a given standard sequence, the orientation of the standard sequence is used for all segments of the standard sequence for which parallel sequences exist.

**Version**

Change number/ version; for information purposes only.

**Branch operation**

Operation number of a standard sequence operation,

- *before* which a parallel sequence should branch off, or

- *from* which an alternative sequence should be replaced.

This is a mandatory field for parallel and alternative sequences. For a standard sequence, this field must remain empty.

If manually setting up an alternative or parallel sequence, the branch operation of the standard sequence must already exist in the order backlog. When a sequence is handed over via an interface, a valid operation number also must be handed off (there is no validation check).

**Return operation**

Operation number of a standard sequence operation,

- *after* which a parallel sequence should return, or

- *up to* which an alternative sequence should be replaced.

This is a mandatory field for parallel and alternative sequences. For a standard sequence, this field must remain empty.

If manually setting up an alternative or parallel sequence, the branch-off operation of the standard sequence must already exist in the orders backlog. When a sequence is handed over via an interface, a valid operation number also must be handed off (there is no validation check).

**Reference Sequence**

The reference sequence determines the sequence in the order that the reference operations (branch and return) refer to. This is *always* the standard sequence (sequence number 0).

This is a mandatory field for parallel and alternative sequences. The standard sequence must already exist.

For a standard sequence, this field must remain empty.

## Toolbar

**Activate**

Activate an alternative sequence

**Deactivate**

Deactivate an alternative sequence

**Edit orders**

Calls up the application Edit orders.